

Topic 1.6 - Polynomial Functions and End Behavior

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Determine end behavior from degree parity and leading coefficient.
- Express end behavior with words, arrows, and limit notation and justify leading-term dominance.
- Analyze parameter families in which the degree and end behavior change through cancellation.

Section A: Foundation and representation

Question 1

Foundation

No calculator

State both end-behavior limits of $f(x) = 6x^7 - 2x^4 + 100$.

Question 2

Parity

No calculator

State both end-behavior limits of $g(x) = -0.4x^{10} + 90x^9$.

Section A continued

Question 3

Inverse reasoning

No calculator

A polynomial falls on both ends. What can be concluded about degree parity and leading coefficient?

Question 4

Inverse reasoning

No calculator

A polynomial rises left and falls right. What can be concluded?

Question 5

Dominance

No calculator

Justify the end behavior of $x^6 - 10^8x^5$ algebraically, not from coefficient size.

Section B: Application and reasoning**Question 6**

Parameter

No calculator

For $P_k = (k + 4)x^4 - (k - 1)x^3 + 2$, classify $k = -4$ and $k \neq -4$.

Question 7

Parameter

No calculator

Find a so $(a - 2)x^8 + (3a + 1)x^7 + x$ has opposite end directions.

Question 8

Structure

No calculator

Can a degree-6 polynomial with positive leading coefficient have four local maxima and three local minima? Explain.

Section C: Challenge and error analysis**Question 9**

Error analysis

No calculator

A graph appears down/down in $[-10, 10]$ for $P = x^4 - 10000x^2$. Diagnose the conflict with algebra.

Question 10

Construction

No calculator

Construct a degree-5 polynomial with right end down, zeros $-1, 2$, and $P(0) = 8$.

Question 11

Context

No calculator

A profit model is valid only for $0 \leq x \leq 40$ but has $P(x) \rightarrow \infty$ as $x \rightarrow \infty$. Explain why the limit need not predict actual unlimited profit.

Question 12

Challenge

No calculator

Determine all k for which $(k^2 - 1)x^6 + (k - 1)x^5 - x^4$ has both ends down.

Section D: AP-style multiple choice

MCQ 1

No calculator

Which end behavior matches $-7x^{12} + x^{20}$?

- A. up/up
- B. down/down
- C. down/up
- D. up/down

MCQ 2

No calculator

A polynomial has left end up and right end down. Which leading term is possible?

- A. $4x^8$
- B. $-4x^8$
- C. $4x^9$
- D. $-4x^9$

MCQ 3

No calculator

For $P_a = (a - 1)x^4 + 3x^3$, when do the ends move oppositely?

- A. $a < 1$
- B. $a = 1$
- C. $a > 1$
- D. All a

MCQ 4

No calculator

Which statement is always true?

- A. A degree-6 polynomial has six real zeros.
- B. A positive even-degree polynomial is increasing everywhere.
- C. Lower-degree terms do not change polynomial end behavior.
- D. An odd-degree polynomial has two upward ends.

AP-style free response 1**No calculator - 6 points**

Let $P_a(x) = (a - 3)x^6 + (2a + 1)x^5 - 7x + 4$.

A. Classify end behavior for $a > 3$.

B. Classify end behavior for $a < 3$.

C. Determine end behavior for $a = 3$.

D. Find all a for which ends move oppositely and justify.

AP-style free response 2**Calculator permitted - 6 points**

Profit, in thousands of QAR, is modeled on $0 \leq t \leq 8$ by $R(t) = -0.5t^4 + 6t^3 - 18t^2 + 40$.

A. Use technology to find all local extrema on the domain.

B. Determine absolute maximum and minimum on the domain.

C. State the polynomial's global end behavior.

D. Explain why global end behavior does not describe profit beyond the stated domain.
